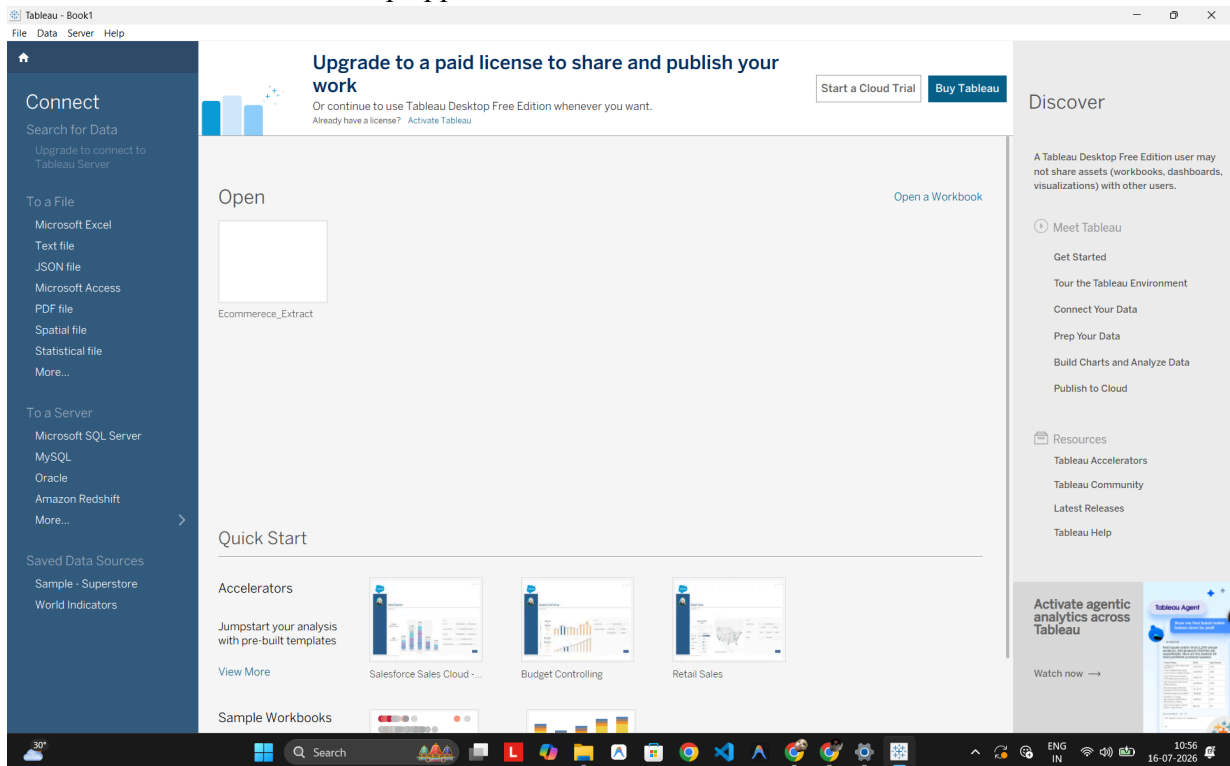


Sheth L.U.J & Sir M.V College
DATA VISUALIZATION WITH POWER BI AND TABLEAU

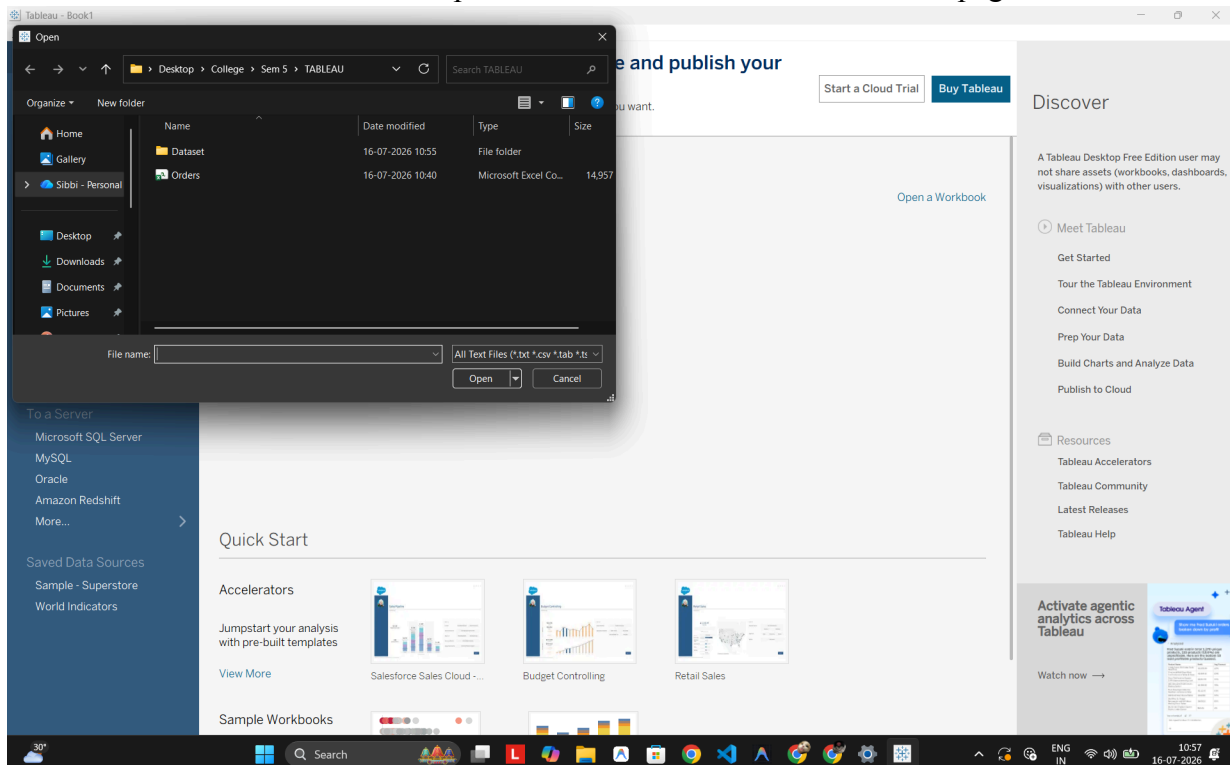
Lesson 2: Working with Data Sources and Data Preparation in Tableau

Step A: Connecting to Different Data Sources

1. Launch the Tableau Desktop application.



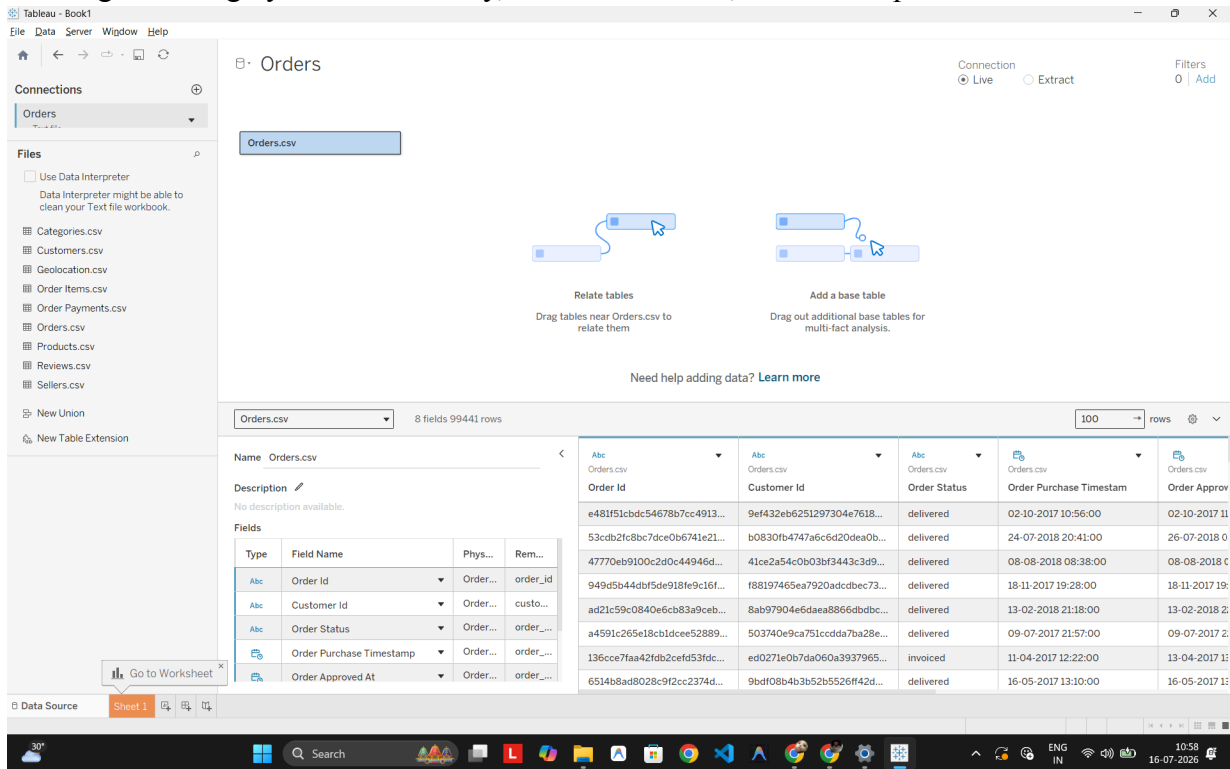
2. Locate the blue Connect control panel on the far-left section of the start page.



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DATA VISUALIZATION WITH POWER BI AND TABLEAU

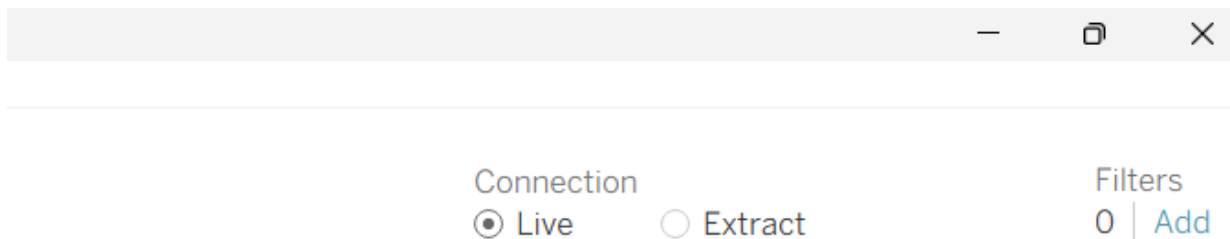
3. Under the To a File sub-header, click on Text File (Tableau classifies .csv spreadsheet extensions as flat text files).

4. Navigate through your file directory, select Orders.csv, and click Open.



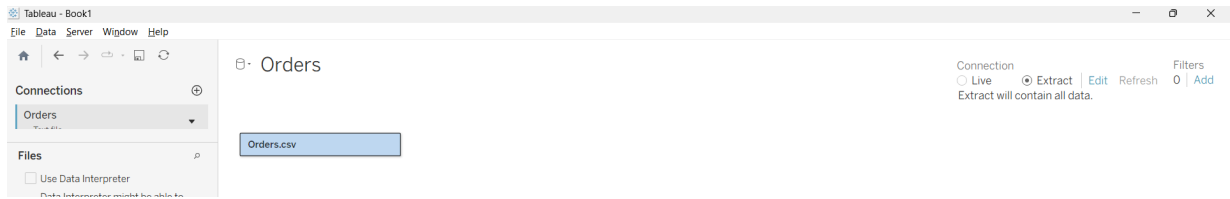
Step B: Creating Data Extracts vs. Live Connections

1. Once the dataset populates the main staging view, navigate to the top-right corner of the Data Source interface.

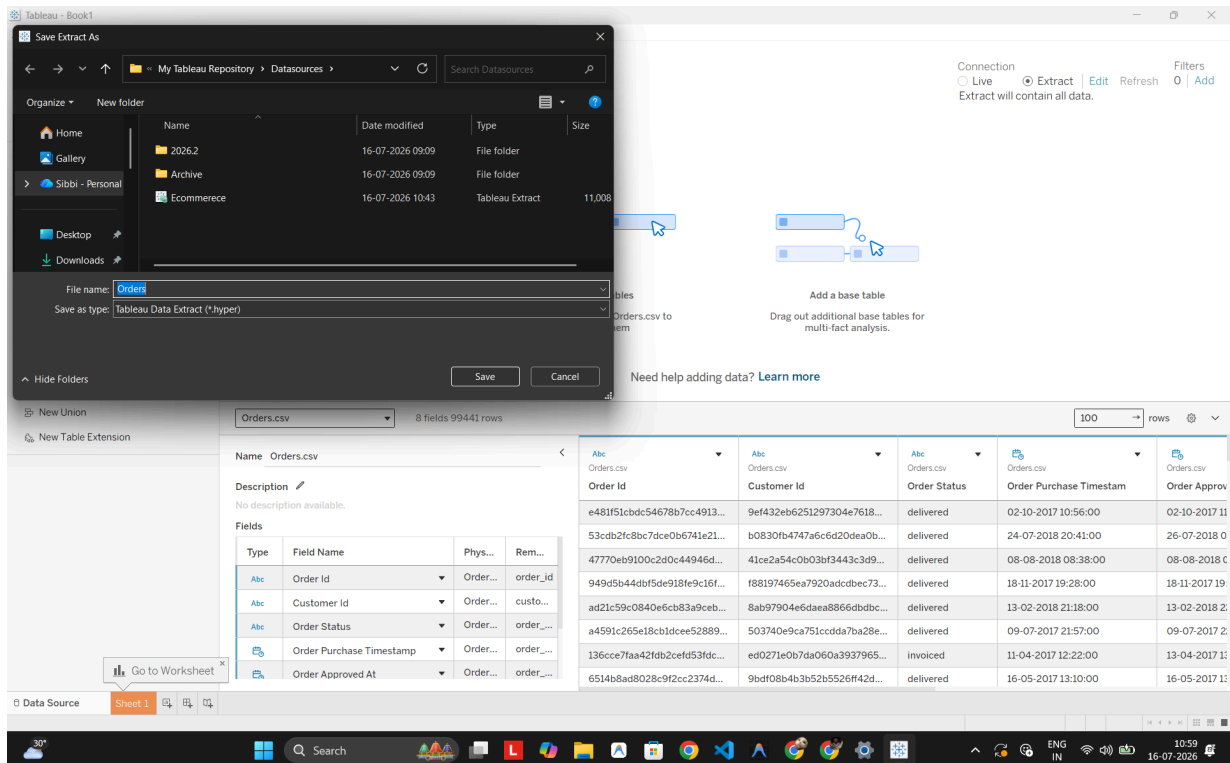


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DATA VISUALIZATION WITH POWER BI AND TABLEAU

2. Locate the connection mode toggle section.
3. Switch the active selection from Live over to Extract.

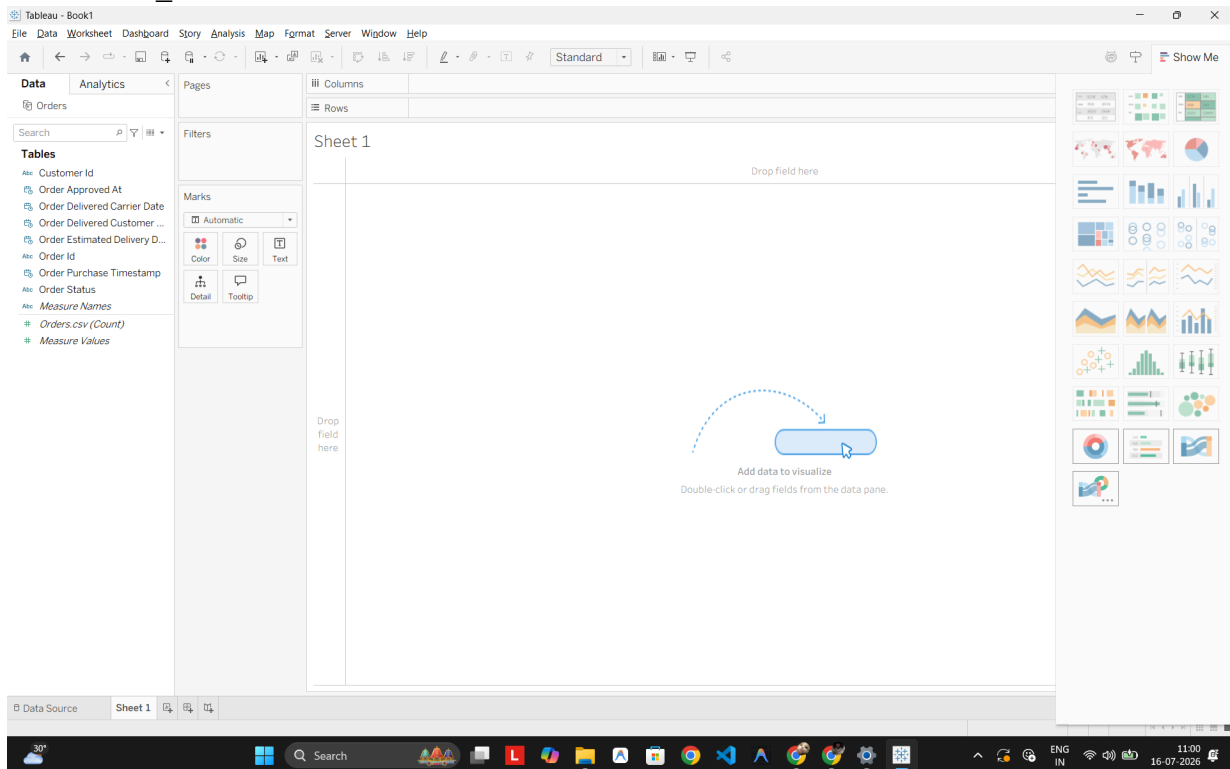


4. Click on the Sheet 1 navigation tab located at the bottom-left corner of the workspace screen.



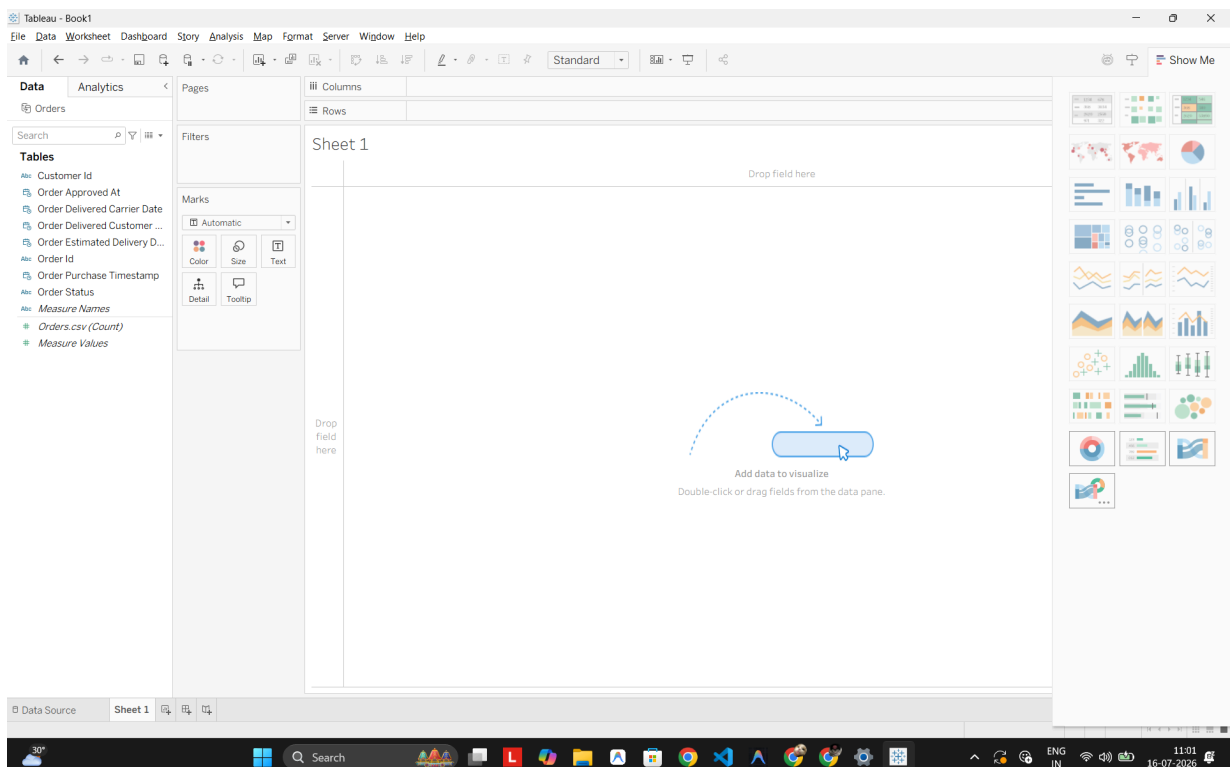
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DATA VISUALIZATION WITH POWER BI AND TABLEAU

5. A system dialog save window will immediately appear. Name the local snapshot file Ecommerce_Extract and click Save.

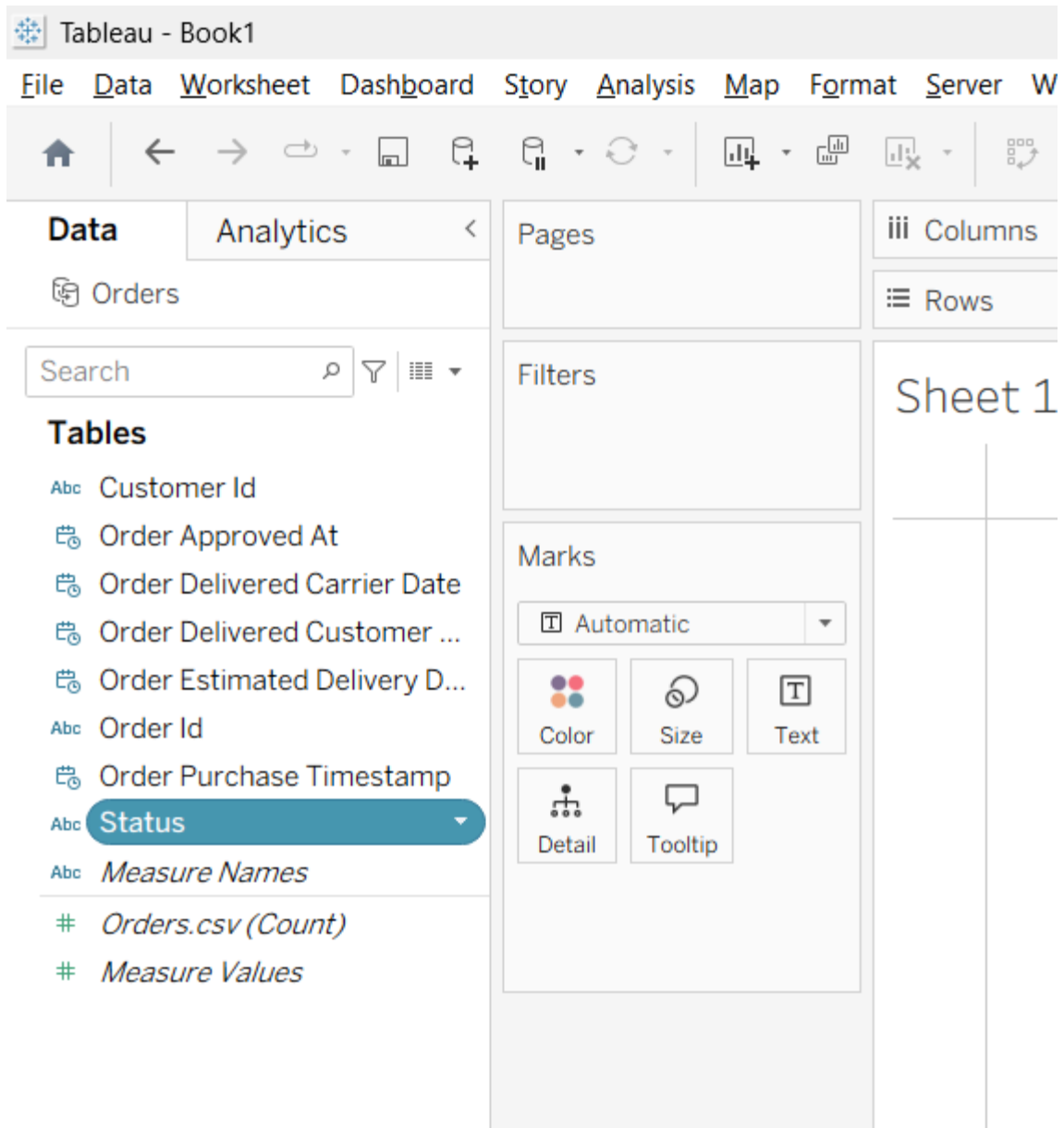


Step C: Managing Operational Metadata

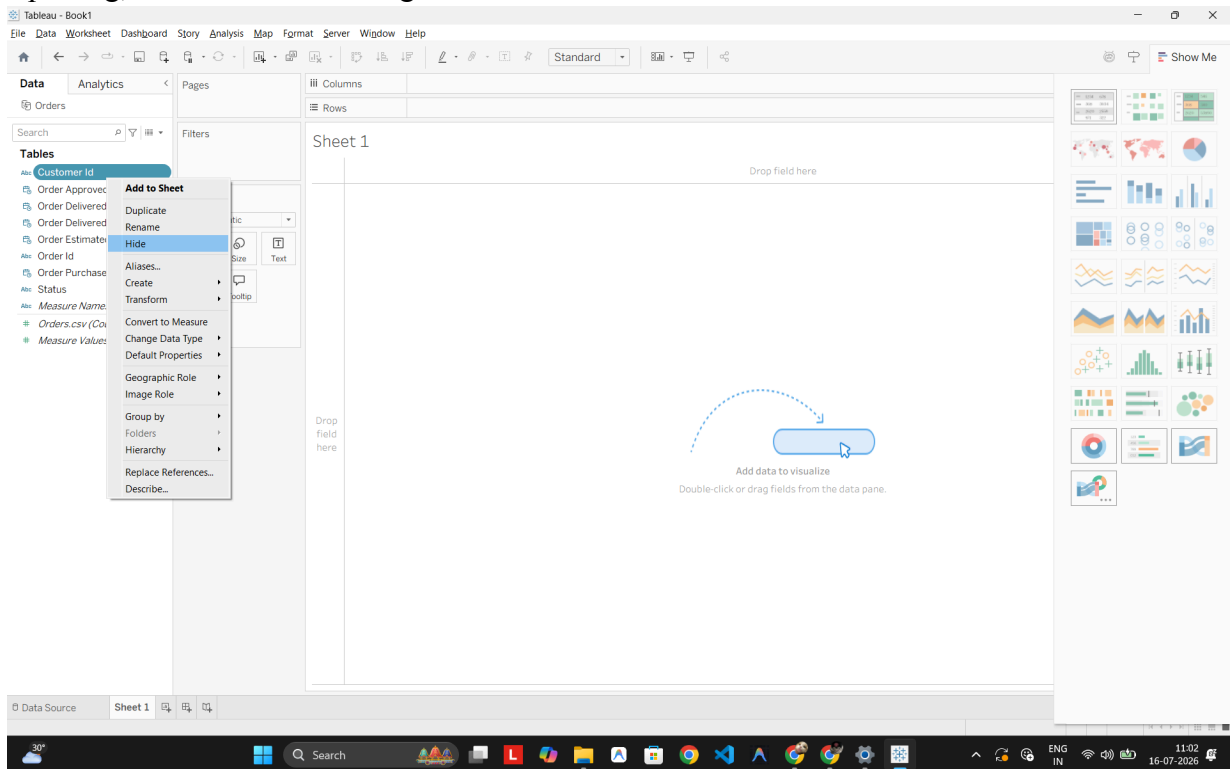
1. Navigate to the blank workspace canvas on Sheet 1 and view the left-hand side Data Pane.



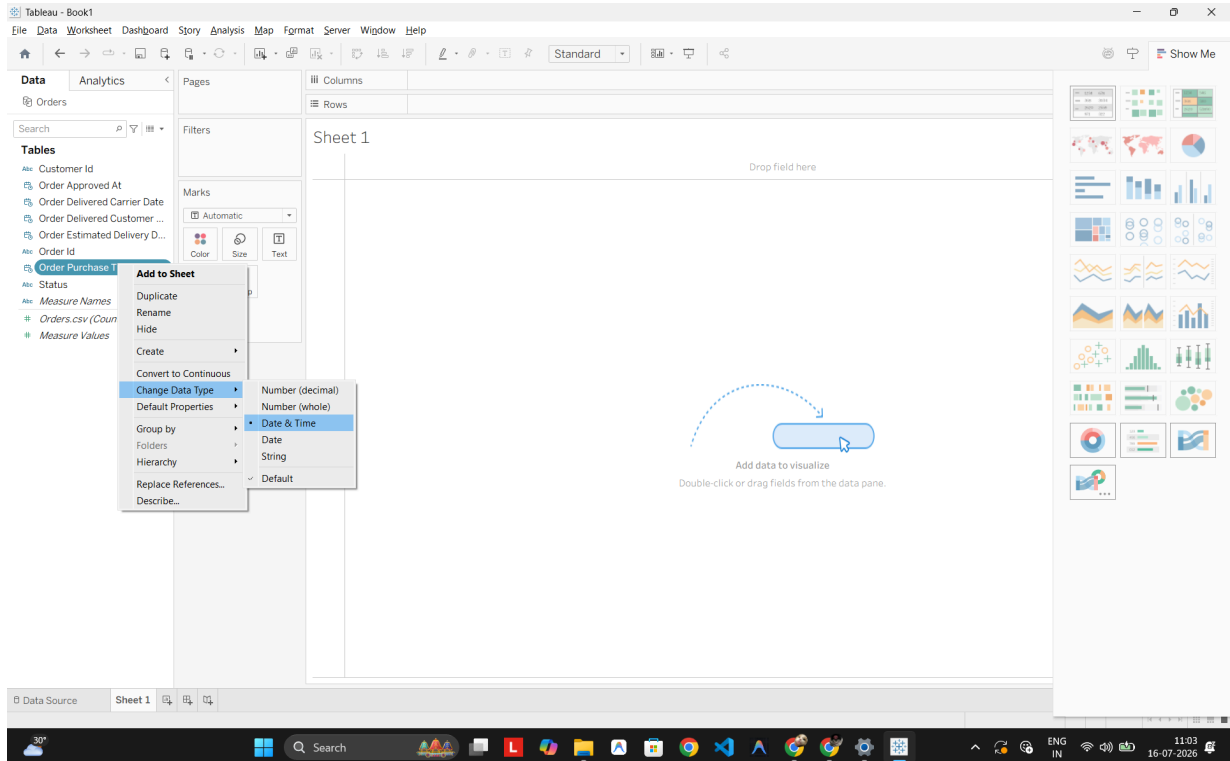
2. Rename a Field: Locate the dimensional attribute field labeled order_status or Order Status. Right-click the field name, select Rename, type in Status, and hit Enter.



3. Hide a Field: Locate an internal tracking string variable that will not be used in visual reporting, such as customer. Right-click the field name and select Hide.



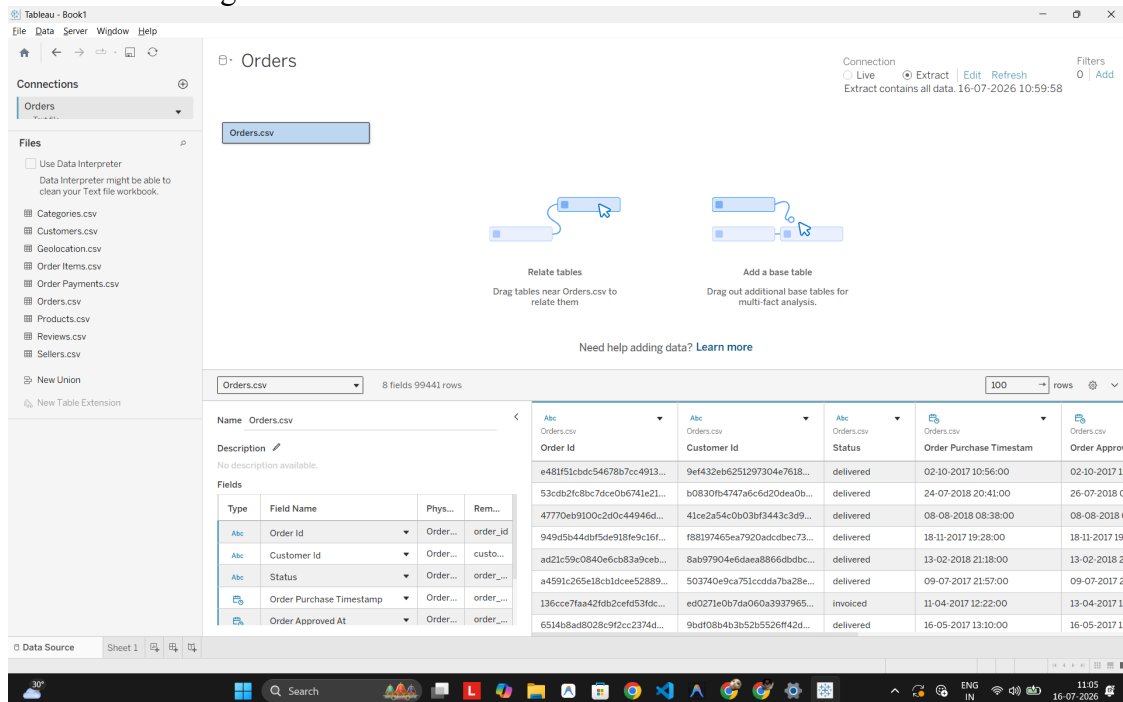
4. Change a Data Type: Locate the operational timestamp column labeled `order_purchase_timestamp`. Click directly on the data type icon (which may incorrectly display as text `Abc` or a calendar date) positioned immediately to the left of the field name. Select **Date & Time** from the context dropdown menu to force proper structural recognition (Note: It may by default selected to Date and Time).



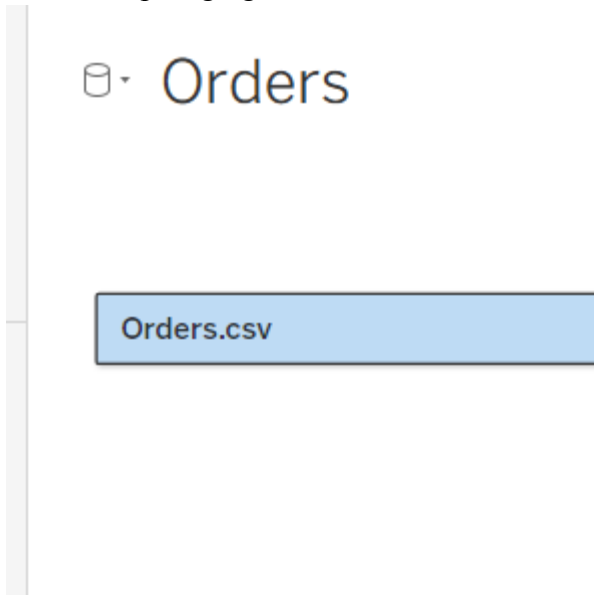
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DATA VISUALIZATION WITH POWER BI AND TABLEAU

Step D: Performing Relational Joins on Multiple Tables

1. Click the Data Source tab positioned at the bottom-left corner of the window to return to the data schema designer.



2. In the top staging canvas area, locate the rectangular Orders.csv database block.



3. Double-click directly on the Orders.csv block. This action drills down through the logical relationship layer and enters the Physical Layer canvas space.

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DATA VISUALIZATION WITH POWER BI AND TABLEAU

Tableau - Book1

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Categories.csv

Customers.csv

Geolocation.csv

Order Items.csv

Order Payments.csv

Orders.csv

Products.csv

Reviews.csv

Sellers.csv

New Union

New Table Extension

Orders

Connection

Live Extract Edit Refresh

Extract contains all data. 16-07-2026 10:59:58

Filters

0 Add

Orders.csv is made of 1 table.

Orders.csv

Orders.csv

8 fields 99441 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	Status	Order...	order_...
Abc	Order Purchase Timestamp	Order...	order_...
Abc	Order Approved At	Order...	order_...

Order Id	Customer Id	Status	Order Purchase Timestamp	Order Approved At
e481f51c8dc54678b7cc4913...	9ef432eb6251297304e7618...	delivered	02-10-2017 10:56:00	02-10-2017 11...
53c8b2f8b8c7dce0b6741e21...	b0830fb4747a6c6d20dea0b...	delivered	24-07-2018 20:41:00	26-07-2018 0...
47770eb9100c2d0c44946d...	41ce2a54c0b03bf3443c3d9...	delivered	08-08-2018 08:38:00	08-08-2018 0...
949d5b44dbf5de918fe9c16f...	f88197465ea7920adcddec73...	delivered	18-11-2017 19:28:00	18-11-2017 19...
ad21c59c0840e6cb83a9ceb...	8ab97904e6daea866dbdbc...	delivered	13-02-2018 21:18:00	13-02-2018 2...
a4591c265e18cb1dcee52889...	503740e9ca751ccdda7ba28e...	delivered	09-07-2017 21:57:00	09-07-2017 2...
136cce7faa42fdb2cfd53fdc...	ed0271e0b7da060a3937965...	invoiced	11-04-2017 12:22:00	13-04-2017 1...
6514b8ad8028c9f2cc2374d...	9bd08b4b3b52b5526f42d...	delivered	16-05-2017 13:10:00	16-05-2017 1...

Data Source Sheet 1

4. From the left-hand files listing pane, click and drag the Order Payments.csv file, dropping it directly next to the active Orders.csv block. A traditional Venn diagram icon link will snap open between them.

Tableau - Book1

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter

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Categories.csv

Customers.csv

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Order Items.csv

Order Payments.csv

Orders.csv

Products.csv

Reviews.csv

Sellers.csv

New Union

New Table Extension

Orders

Connection

Live Extract Edit Refresh

Extract will contain all data.

Filters

0 Add

Orders.csv is made of 2 tables.

Order Payments.csv

Join

Inner Left Right Full Outer

Date Source

Order Payments.csv

Order Id

order_id (Order ...)

Add new join clause

Orders.csv

10 fields 122978 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

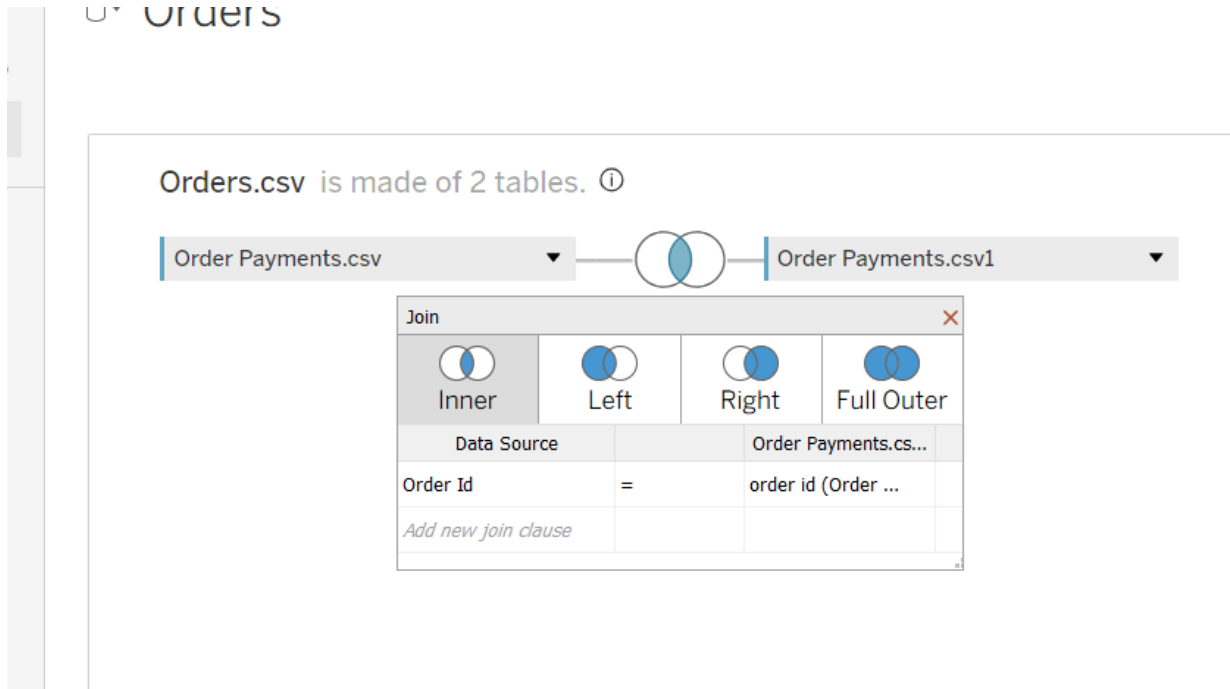
Type	Field Name	Phys...	Rem...
Abc	Order Id	Order ...	order_id
#	payment_sequential	Order ...	paym...
Abc	payment_type	Order ...	paym...
#	payment_installments	Order ...	paym...
#	payment_value	Order ...	paym...

Order Id	payment_sequential	payment_type	payment_installments	payment_value
b81ef226f3fe1789b1e8b2aca...	1	credit_card	8	99.33
a9810da82917af2d9aefcd127...	1	credit_card	1	24.39
25e8ea4e93396b6fa0d3dd7...	1	credit_card	1	65.71
ba78997921bbdc1373bb41e...	1	credit_card	8	107.78
42fd880ba16b47b59251dd4...	1	credit_card	2	128.45
298fcd1f73eb413e4426d01...	1	credit_card	2	96.12
771ee386b00106208a7419e...	1	credit_card	1	81.16
3d7239c394a212faae122962...	1	credit_card	3	51.84

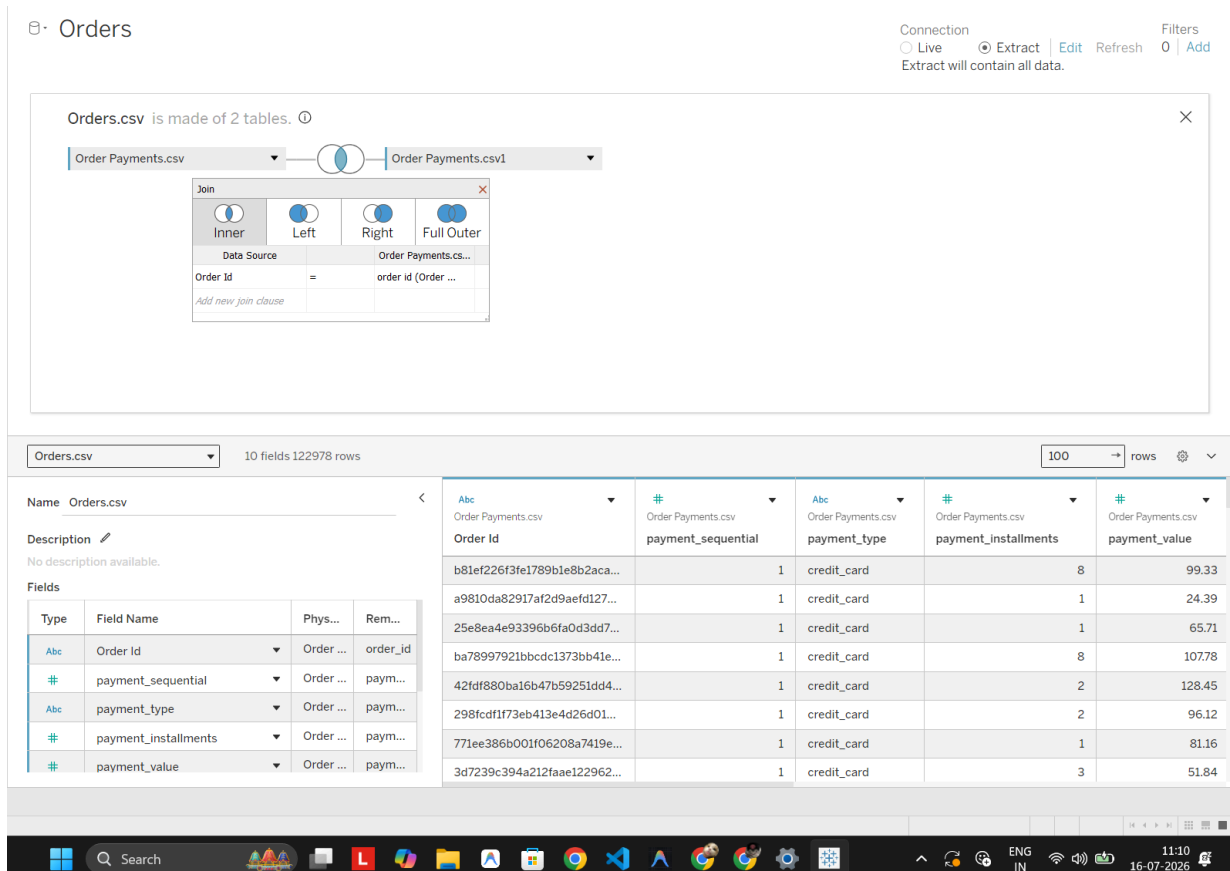
Data Source Sheet 1

5. Click directly on the Venn diagram icon to open the Join configuration properties box.

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DATA VISUALIZATION WITH POWER BI AND TABLEAU



6. Under the data grid coupling properties, select order_id as the matching data key column drop-down for both the left and right tables (Note: in some case Tableau automatically selects the Order Id).



7. Toggle Join Configurations:

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DATA VISUALIZATION WITH POWER BI AND TABLEAU

o Click Inner: Observe the underlying data preview window at the bottom. The rows instantly filter to show only records that contain both an order entry and a corresponding transaction payment receipt.

Tableau - Book1

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Categories.csv

Customers.csv

Geolocation.csv

Order Items.csv

Order Payments.csv

Orders.csv

Products.csv

Reviews.csv

Sellers.csv

New Union

New Table Extension

Orders

Orders.csv is made of 2 tables.

Order Payments.csv

Order Payments.csv1

Join

Inner Left Right Full Outer

Data Source

Order Payments.csv1

Order Id

order id (Order ...)

Add new join clause

Orders.csv

10 fields 122978 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order ...	order_id
#	payment_sequential	Order ...	paym...
Abc	payment_type	Order ...	paym...
#	payment_installments	Order ...	paym...
#	payment_value	Order ...	paym...

Order Payments.csv	#	Order Payments.csv	#	Order Payments.csv	#
Order Id	payment_sequential	payment_type	payment_installments	payment_value	
b81ef226f31e1789b1e8b2aca...	1	credit_card	8	99.33	
a9810da82917af2d9aefd127...	1	credit_card	1	24.39	
25e8ea4e93396b6fa0d3dd7...	1	credit_card	1	65.71	
ba78997921bbdc1373bb41e...	1	credit_card	8	107.78	
42fdf880ba16b47b59251dd4...	1	credit_card	2	128.45	
298fcdf173eb413e4d26d01...	1	credit_card	2	96.12	
771ee386b001f06208a7419e...	1	credit_card	1	81.16	
3d7239c394a212faae122962...	1	credit_card	3	51.84	

Data Source

Sheet 1

30°

Search

ENG IN

11:12

16-07-2026

8. Leave the configuration set to a Left Join to preserve complete order history logs for the remainder of this exercise.

Tableau - Book1

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter

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Order Items.csv

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Orders.csv

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Sellers.csv

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New Table Extension

Orders

Orders.csv is made of 2 tables.

Order Payments.csv

Order Payments.csv1

Join

Inner Left Right Full Outer

Data Source

Order Payments.csv1

Order Id

order id (Order ...)

Add new join clause

Orders.csv

10 fields 122978 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order ...	order_id
#	payment_sequential	Order ...	paym...
Abc	payment_type	Order ...	paym...
#	payment_installments	Order ...	paym...
#	payment_value	Order ...	paym...

Order Payments.csv	#	Order Payments.csv	#	Order Payments.csv	#
Order Id	payment_sequential	payment_type	payment_installments	payment_value	
b81ef226f31e1789b1e8b2aca...	1	credit_card	8	99.33	
a9810da82917af2d9aefd127...	1	credit_card	1	24.39	
25e8ea4e93396b6fa0d3dd7...	1	credit_card	1	65.71	
ba78997921bbdc1373bb41e...	1	credit_card	8	107.78	
42fdf880ba16b47b59251dd4...	1	credit_card	2	128.45	
298fcdf173eb413e4d26d01...	1	credit_card	2	96.12	
771ee386b001f06208a7419e...	1	credit_card	1	81.16	
3d7239c394a212faae122962...	1	credit_card	3	51.84	

Data Source

Sheet 1

30°

Search

ENG IN

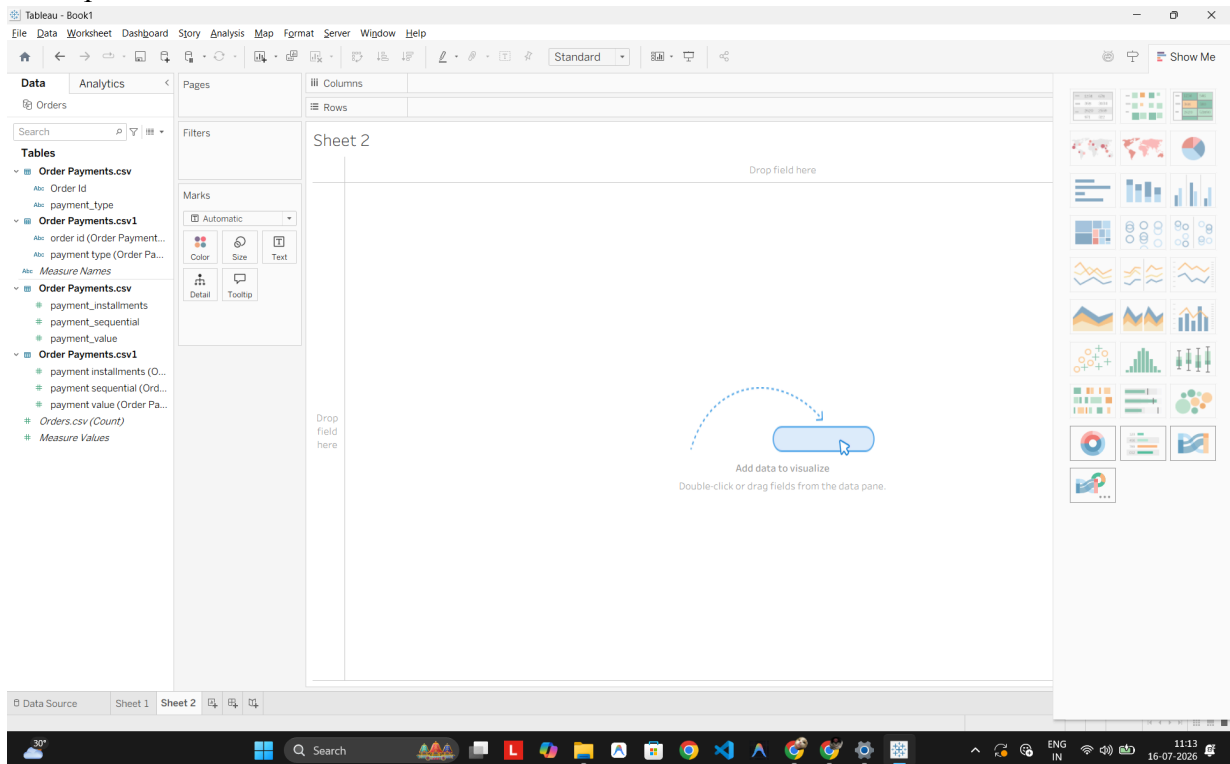
11:12

16-07-2026

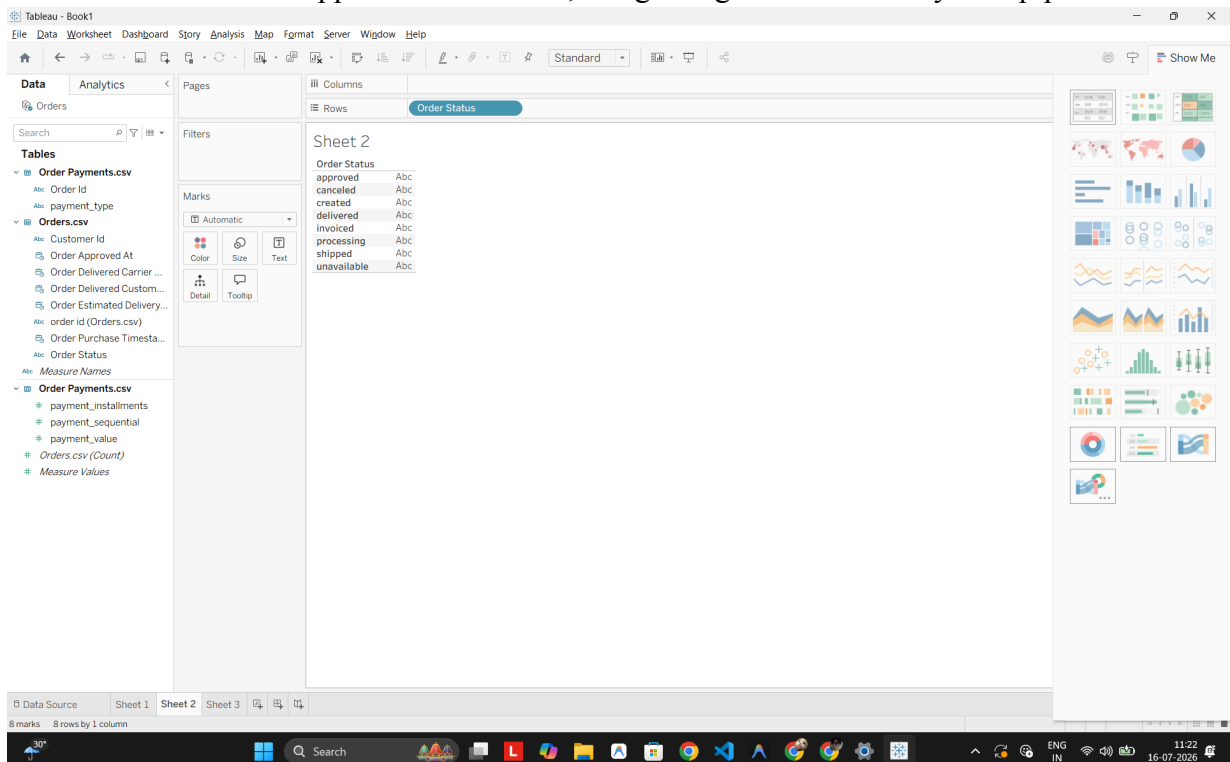
Step E: Applying Data Blending Between Independent Sources

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DATA VISUALIZATION WITH POWER BI AND TABLEAU

1. Create a brand new blank workspace sheet by clicking on the New Worksheet icon (the chart block containing a \$\$ symbol) positioned at the bottom navigation tab area. This opens Sheet 2.



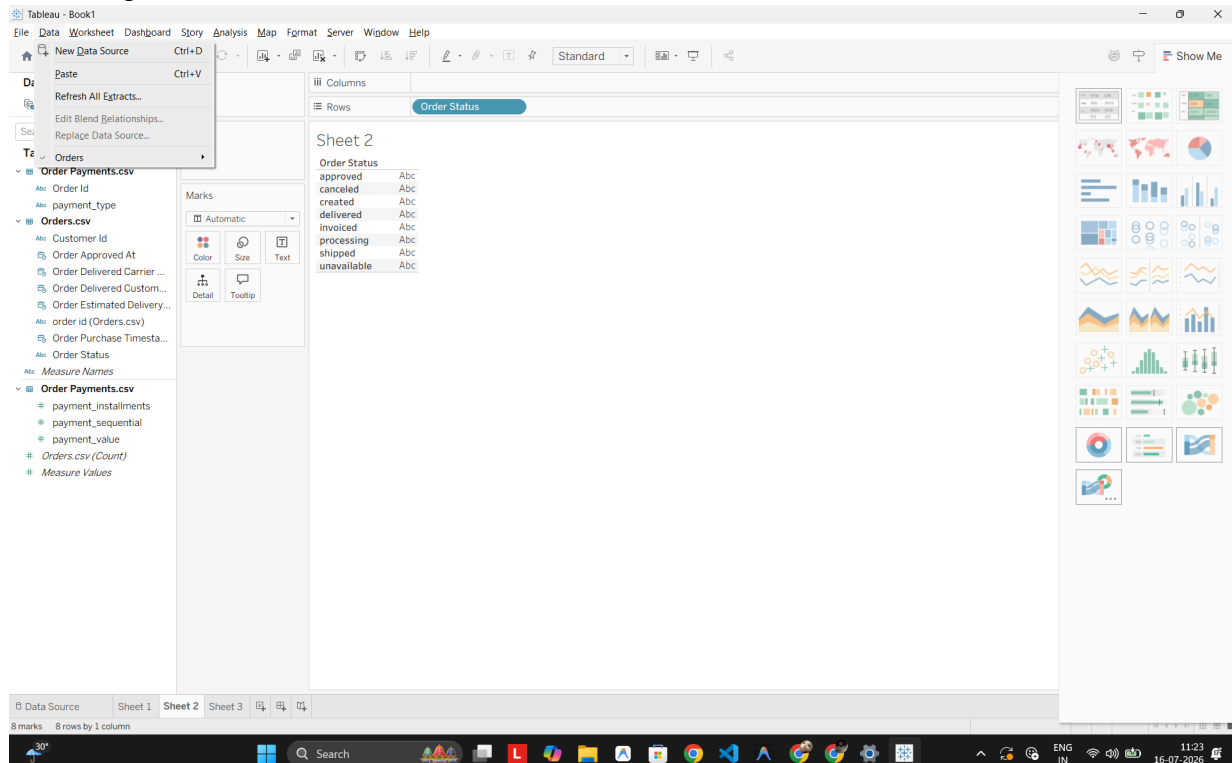
2. Ensure your joined Orders dataset connection is highlighted at the top left of the Data Pane. Drag the newly modified Status dimension field and drop it onto the Rows shelf. A blue checkmark will appear on the source, designating it as the Primary data pipeline.



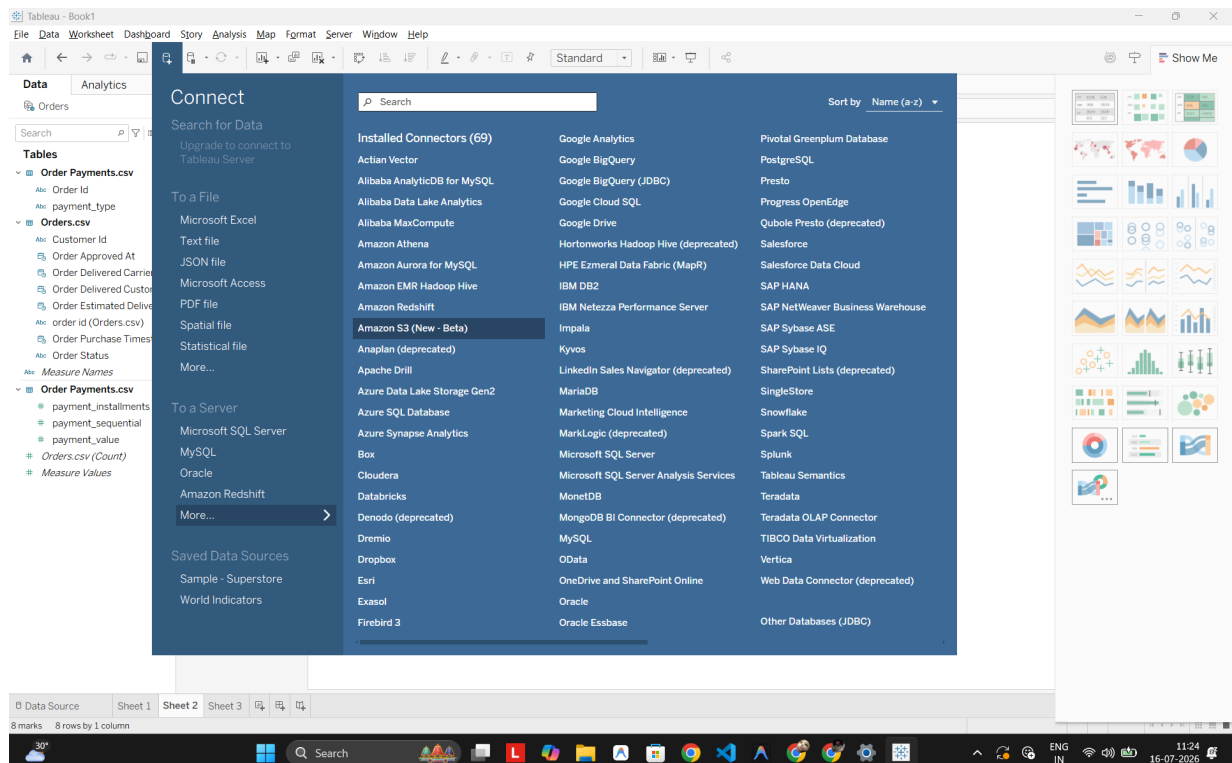
3. Move your mouse to the top application menu bar and click Data > Edit Blend

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DATA VISUALIZATION WITH POWER BI AND TABLEAU

Relationships. If the Option is grayed out as shown in the snapshot,click the New Data Source Option



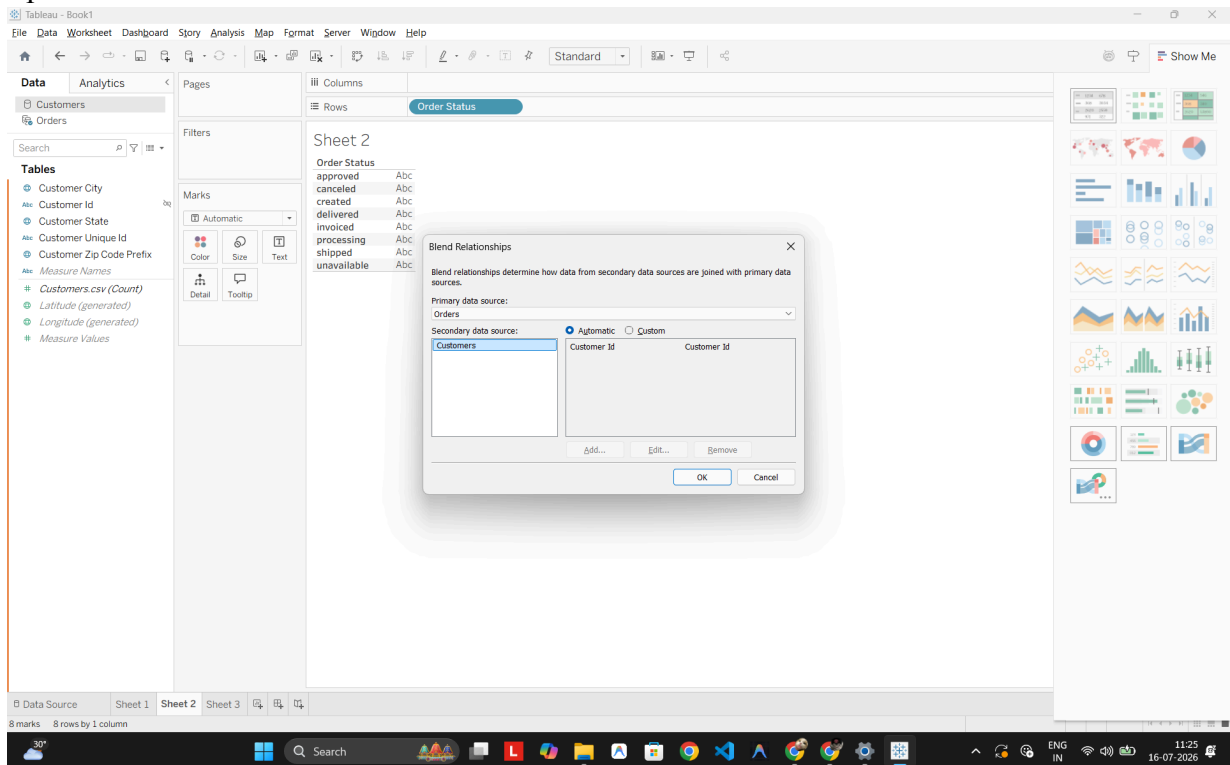
After Clicking new data source you are directed to blue menu window select text file to add the Customers.csv file.



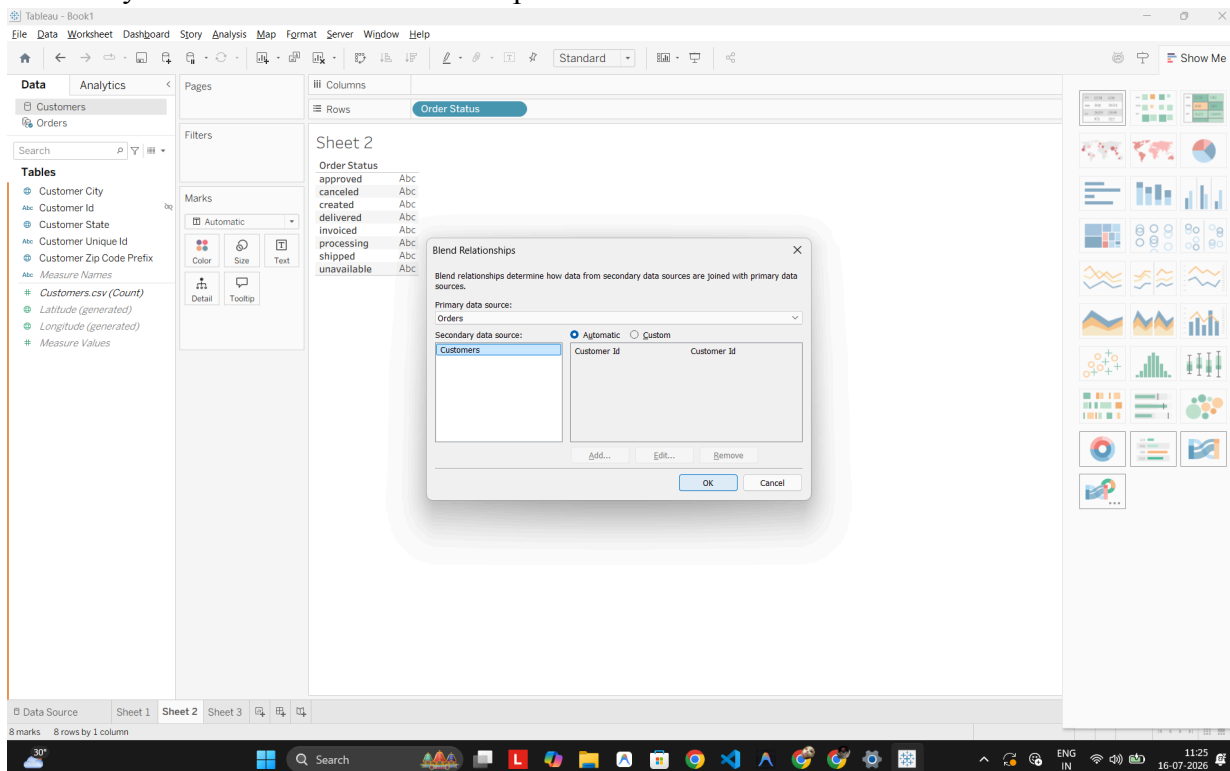
After adding the CSV file get back to sheet 2 and recheck the Edit Blend Relationship

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DATA VISUALIZATION WITH POWER BI AND TABLEAU

Option and click on it



4. Set the Primary data source field box to your active Orders connection, and map the Secondary data source block to the independent Customers dataset file.



5. Switch the selection toggle switch from Automatic over to Custom, click the Add... button on the right, pair customer_id from the left dataset column to customer_id in the right column, and click OK.

Now Navigate Back to data source and change the connection to “LIVE” from “EXTRACT”

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DATA VISUALIZATION WITH POWER BI AND TABLEAU

Tableau - Book1

File Data Server Window Help

Connections

Customers

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Customers

Customers.csv

Connection: Live Extract Filters: 0 Add

Relate tables
Drag tables near Customers.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Customers.csv 5 fields 99441 rows 100 rows

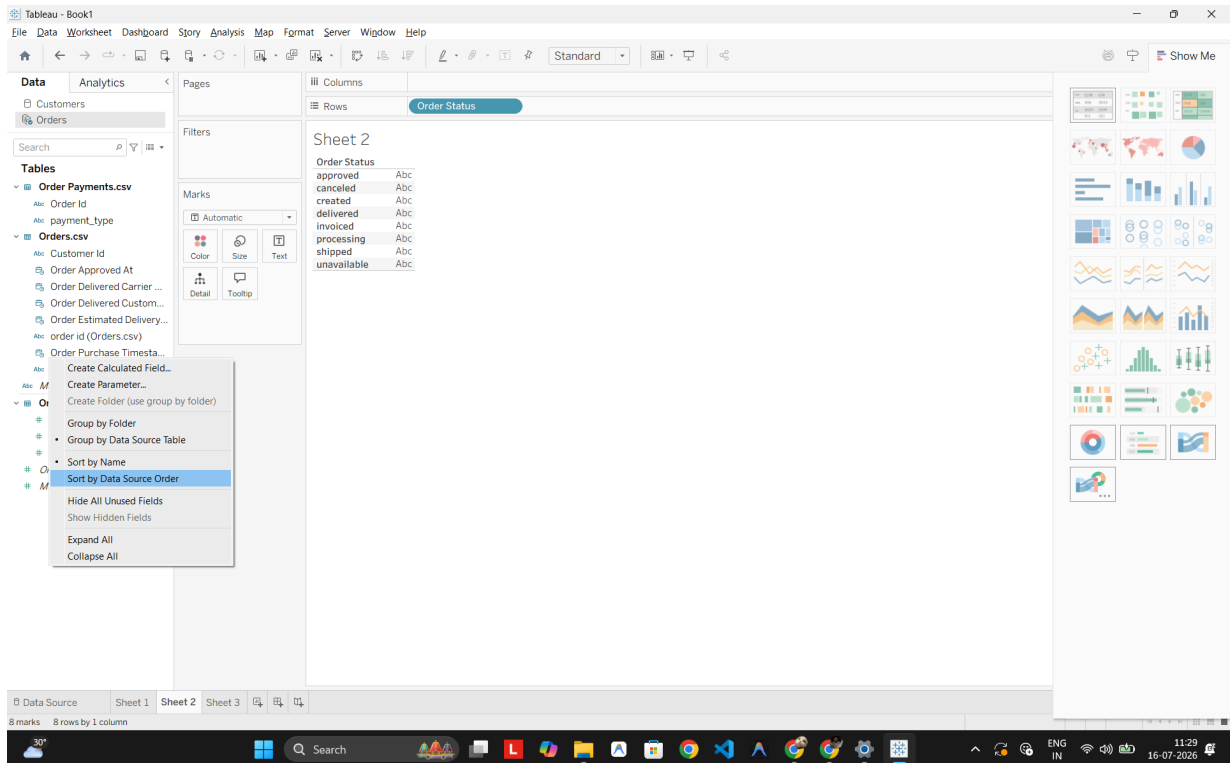
Name	Customers.csv	Customers.csv	Customers.csv	Customers.csv	Customers.csv
Description					
Fields	Type	Field Name	Physical ...	Rem...	
Abc	Customer Id	Customers...	custo...		
Abc	Customer Unique Id	Customers...	custo...		
	Customer Zip Code Prefix	Customers...	custo...		
	Customer City	Customers...	custo...		
	Customer State	Customers...	custo...		

0 Data Source Sheet 1 Sheet 2 Sheet 3

30° Search ENG IN 11:28 16-07-2026

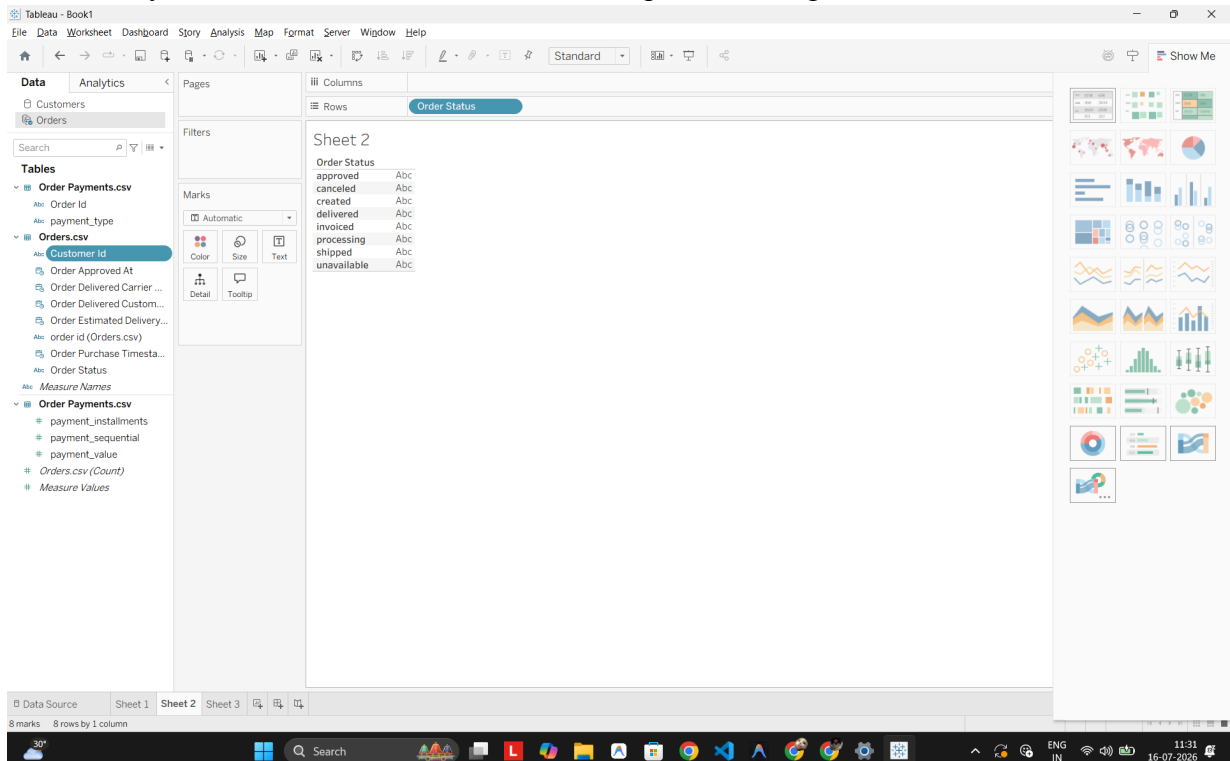
Now Go Back to any to the Sheet 2 and right click on the Data Pane's empty space and click on "Show Hidden Fields"

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DATA VISUALIZATION WITH POWER BI AND TABLEAU

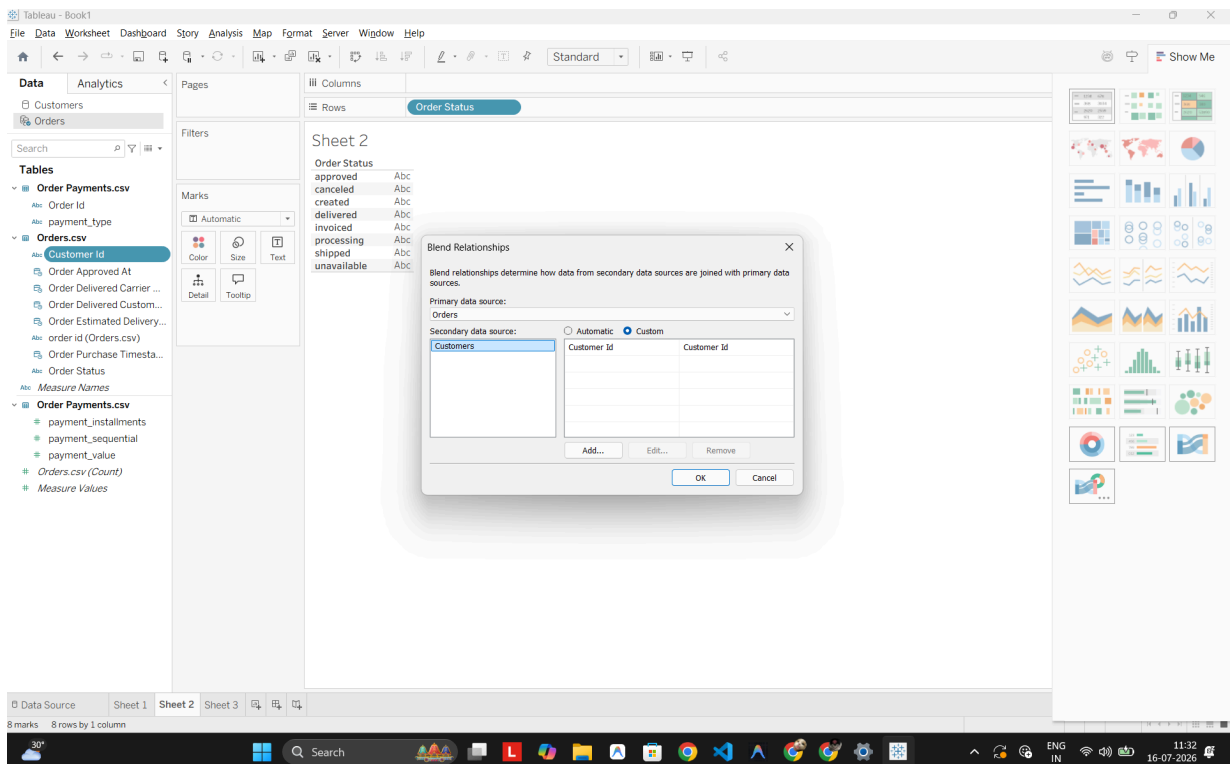


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DATA VISUALIZATION WITH POWER BI AND TABLEAU

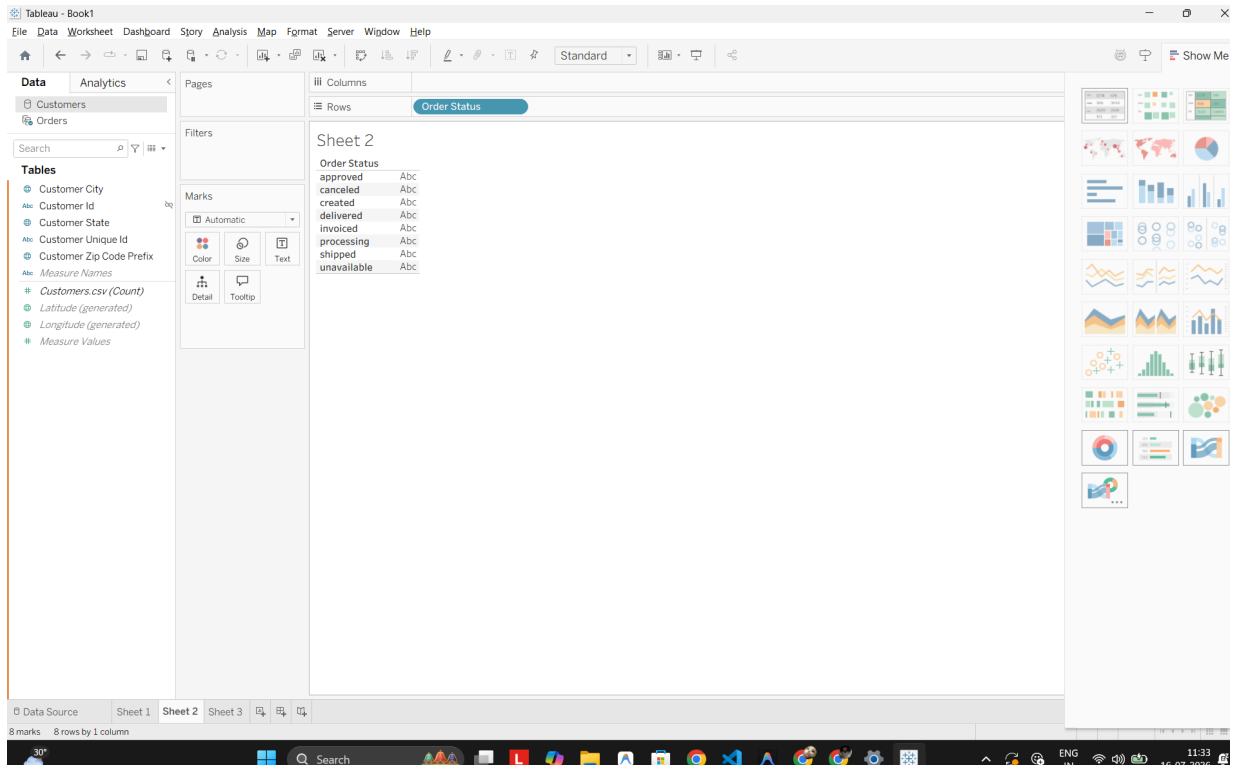
After Clicking it the Customer Id comes back into the data pane and make sure you click the “eye button” next to it marked in bellow provided snapshot



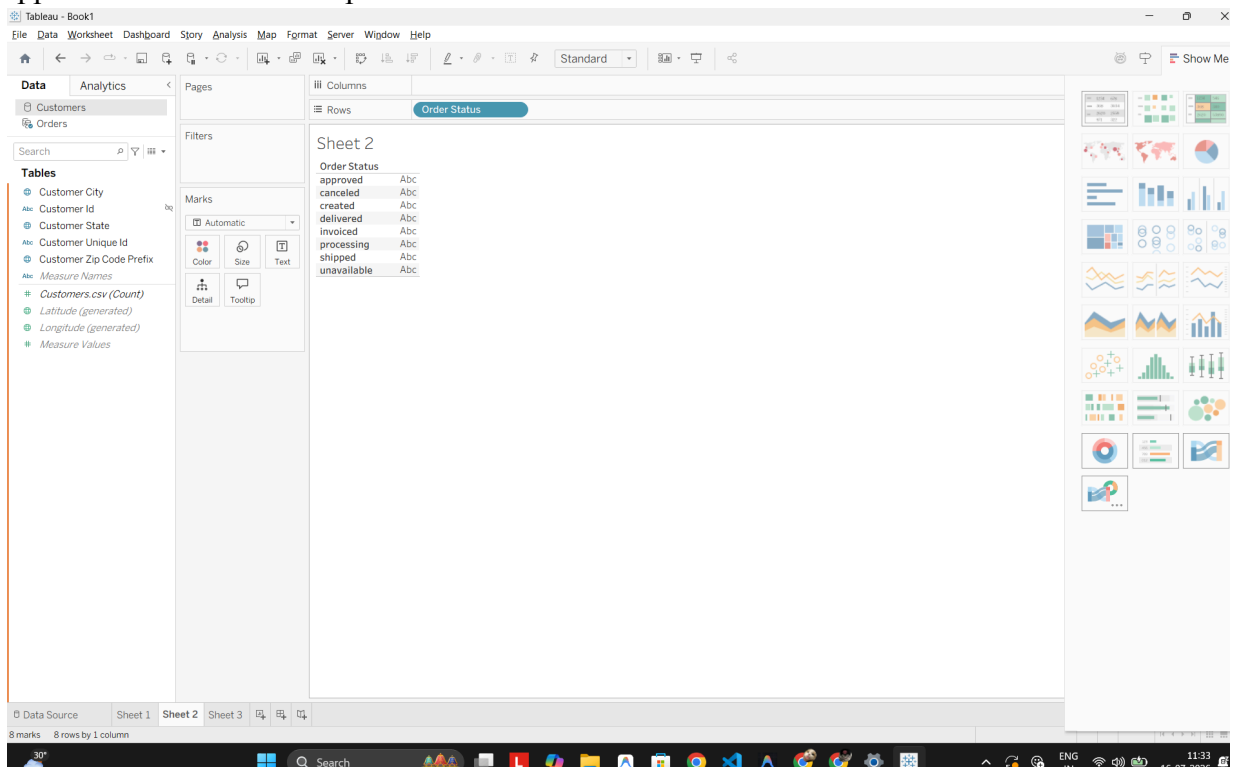
Now go back to Edit Blend Relationship option and after clicking it you may see both Customer Id in Left and Right, After that click on Custom it changes from automatic to custom and click Ok



6. Now, go back to the top-left section of your Data Pane and click directly on the separate Customers dataset tracking label. An orange chain link icon will now appear locked together immediately next to the customer_id field.



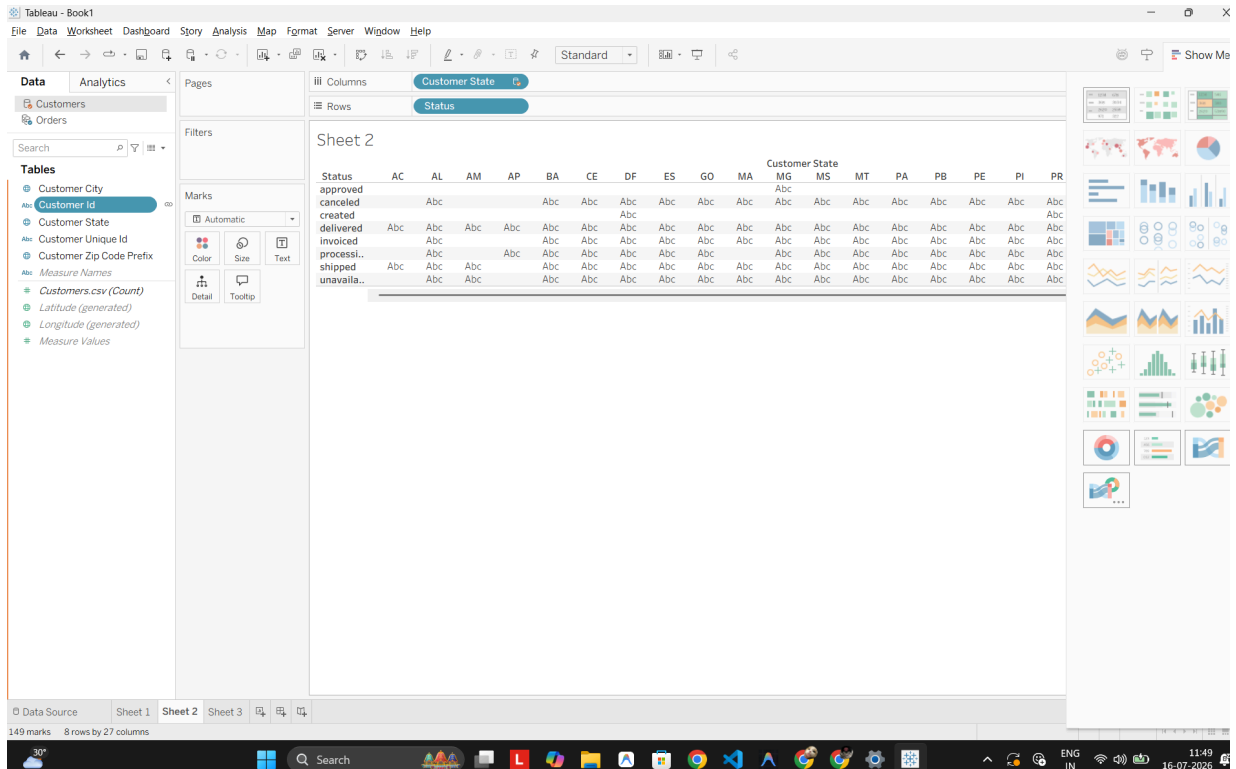
At the start you may not see the orange tick on the Customer database for that click on the unlinked chain next to the customer id on the data pane after clicking the orange tick appear as shown in the snapshot



7. Click and drag the customer_state dimension variable out from this secondary dataset list and drop it onto the Columns shelf.

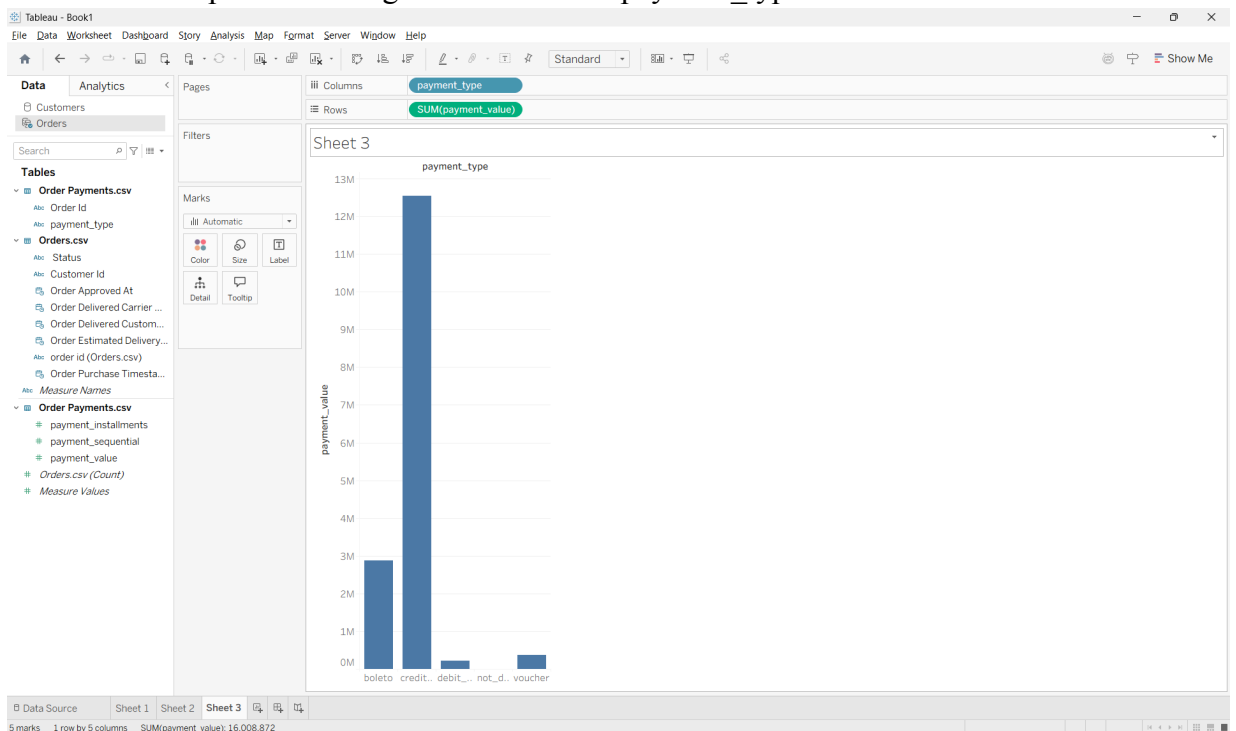
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DATA VISUALIZATION WITH POWER BI AND TABLEAU



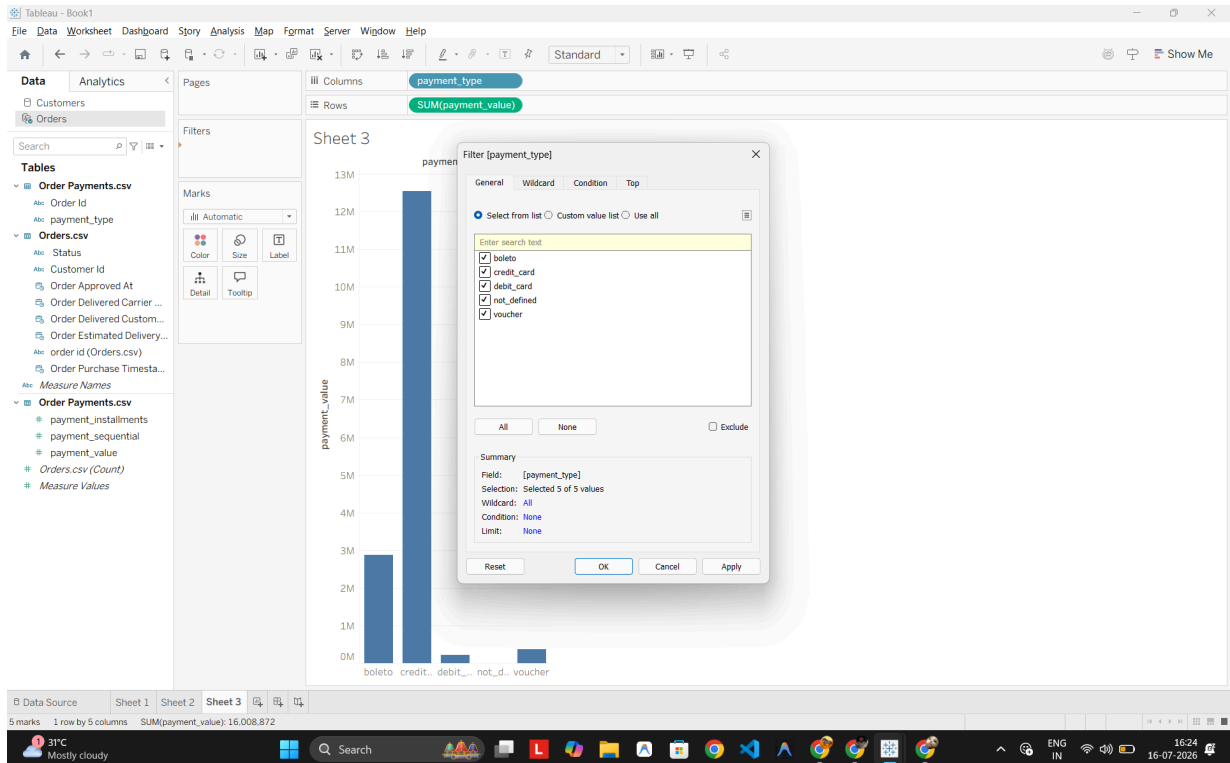
Step F: Implementing Multi-Tiered Filters

1. Open a clean worksheet. Drag the quantitative measure field `payment_value` onto the Rows shelf and place the categorical dimension `payment_type` onto the Columns shelf.



2. Dimension Filter: From your Data Pane list, drag the categorical string attribute `payment_type` and drop it onto the empty Filters shelf positioned immediately above the Marks card layout. Uncheck the box for `not_defined`, check the boxes for `credit_card` and `boleto`, and click OK.

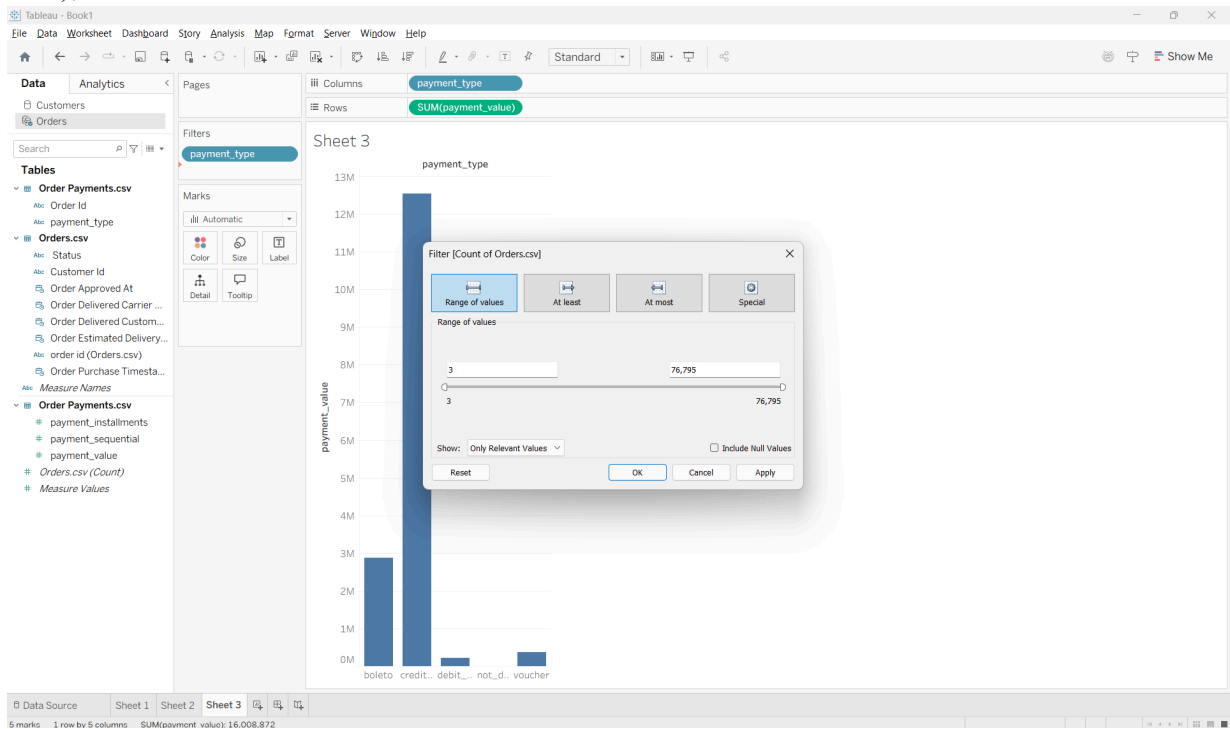
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Measure Filter: Drag the quantitative numeric variable payment_value out from the

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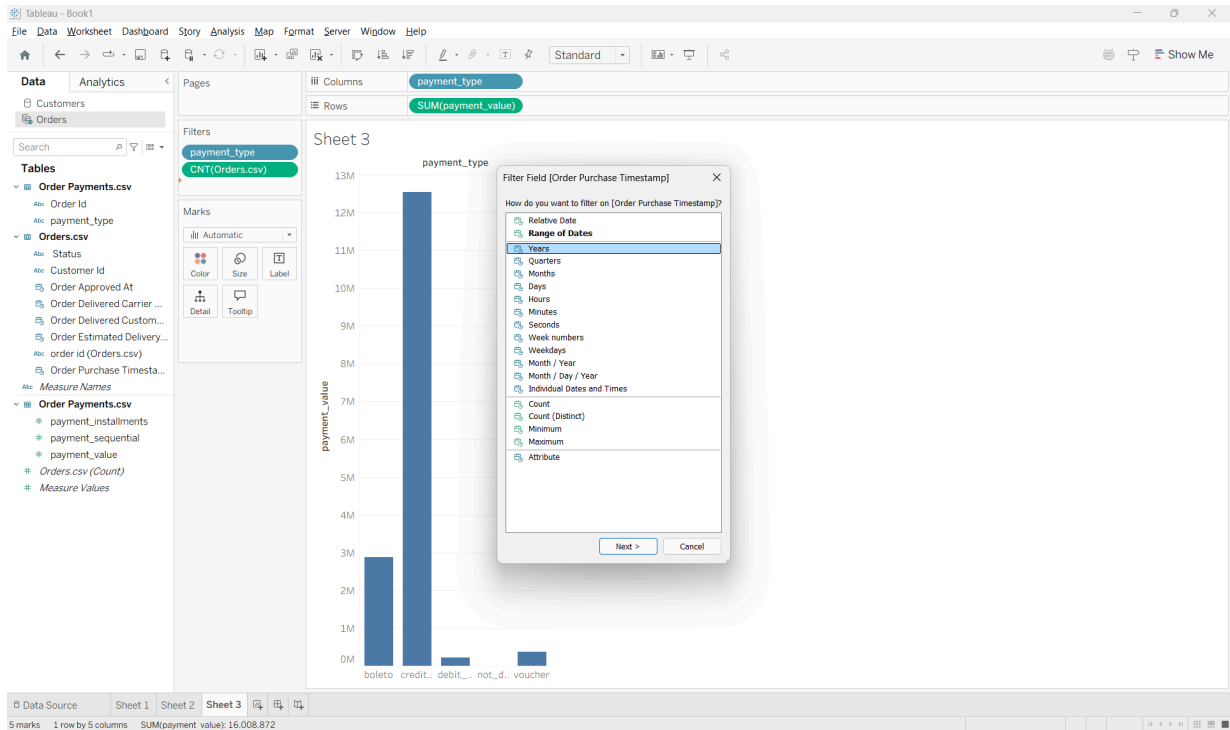
measures list and drop it directly onto the Filters shelf. Select All Values in the prompt window, click Next, select the At Least option condition, type in 50 to screen out microtransactions(it will be at 0.1 at by default Change it by clicking on it and typing 50 on it), and click OK.



4. Date Filter: Drag the calendar configuration time variable `order_purchase_timestamp` and drop it onto the Filters shelf. Select Years or Relative Date from the prompt configuration layout page to isolate the visualization into specific historical reporting boundaries.

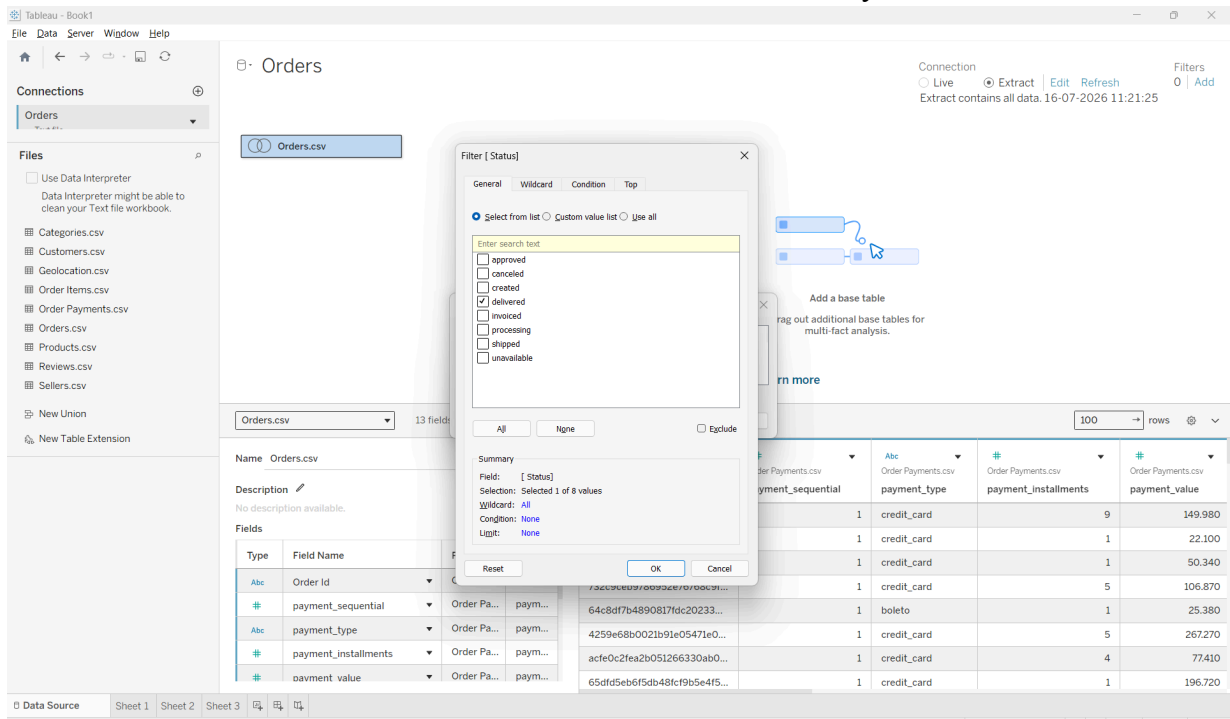
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DATA VISUALIZATION WITH POWER BI AND TABLEAU



Step G: Optimizing Dashboard Performance

1. Extract Filters: Navigate back to the main Data Source configuration workspace screen. Look immediately below the selected Extract radio dial selection button in the upper-right area and click the small Edit link. Click Add, select the categorical field Status, choose to check delivered exclusively, and click OK. Tableau will purge all canceled or unfulfilled order rows from the local extract file entirely.



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DATA VISUALIZATION WITH POWER BI AND TABLEAU

The screenshot shows the Tableau Desktop interface with the 'Orders' data source selected. A filter is applied to the 'Status' field, showing only 'delivered' orders. The 'Edit Data Source Filters' dialog box is open, showing the filter settings. The 'Fields' pane on the left lists the available fields: Order Id, payment_sequential, payment_type, payment_installments, and payment_value. The 'Data Source' pane on the right shows the 'Orders.csv' file with 13 fields and 100756 rows. The 'Data Source' pane also shows the 'Orders.csv' file with 13 fields and 100756 rows.

Tableau - Book1

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Orders

Connection
Live Extract Edit Refresh
Extract contains all data. 16-07-2026 11:21:25

Filters
0 Add

Orders.csv

13 fields

100 rows

Edit Data Source Filters

Filter Scope Details
Status Orders.csv and related L... keeps delivered

Add... Edit... Remove OK Cancel

Fields

Type	Field Name	Physica...	Rem...
ABC	Order Id	Order Pa...	order_id
+	payment_sequential	Order Pa...	paym...
+	payment_type	Order Pa...	paym...
+	payment_installments	Order Pa...	paym...
+	payment_value	Order Pa...	paym...

Data Source Sheet 1 Sheet 2 Sheet 3

Tableau - Book1

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Orders

Connection
Live Extract Edit Refresh
Extract contains all data. 16-07-2026 11:21:25

Filters
1 Edit

Orders.csv

13 fields 100756 rows

100 rows

Fields

Type	Field Name	Phys...	Rem...
ABC	Order Pu	Order...	order_...
ABC	Order Ap	Order...	order_...
ABC	Order De	Order...	order_...
ABC	Order Esi	Order...	order_...

Data Source Sheet 1 Sheet 2 Sheet 3

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DATA VISUALIZATION WITH POWER BI AND TABLEAU

2. Context Filters: Return to an active worksheet layout canvas view. Move your cursor to your Filters shelf, right-click any active filter pill sitting in the shelf area, and select Add to Context from the context dropdown. The selected pill background formatting will immediately change to a dark gray color

